DRUMPANEL · PRECIS

DrumPanel design precis

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DOCUMENT CONTROL

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REVISION HISTORY

Version	Date	Author	Change
0.5	2026-10-24	Amish Chadha	Kit 1.9.1 upgrade: the design is option A of DMP-OPT-001 (proposed, awaiting Amish); STEM fixes F1 to F5 (DMP-DEC-001): first panel of every drum checked for flatness, two drain stands, LEL channel calibrated for heavy vapour, slots 7 to 8 mm, fire-resistant hydraulic fluids refused; throughput 2.20 drums an hour; cost USD 3,566
0.4	2026-10-03	Amish Chadha	Amish's requirement decisions of 2026-10-03 (DMP-DDR-003): lever notching punch, ring head for two ring cuts a pass, raised shear head drive, fine-pitch dial adjuster on the bending roll; figures from DMP-CAL-001 v0.2
0.3	2026-10-03	Amish Chadha	TRL 3; constructable design (DMP-DDR-002): hoop strips cut out so the body gives three panels, slip roll with a pinch pair, shear head on an overhead rail; figures from DMP-CAL-001
0.2	2026-10-03	Amish Chadha	TRL 2; how it works in three stations, key design choices, first-order

			numbers, open questions answered (DMP-DDR-001)
0.1	2026-09-30	Amish Chadha	Initial scaffold

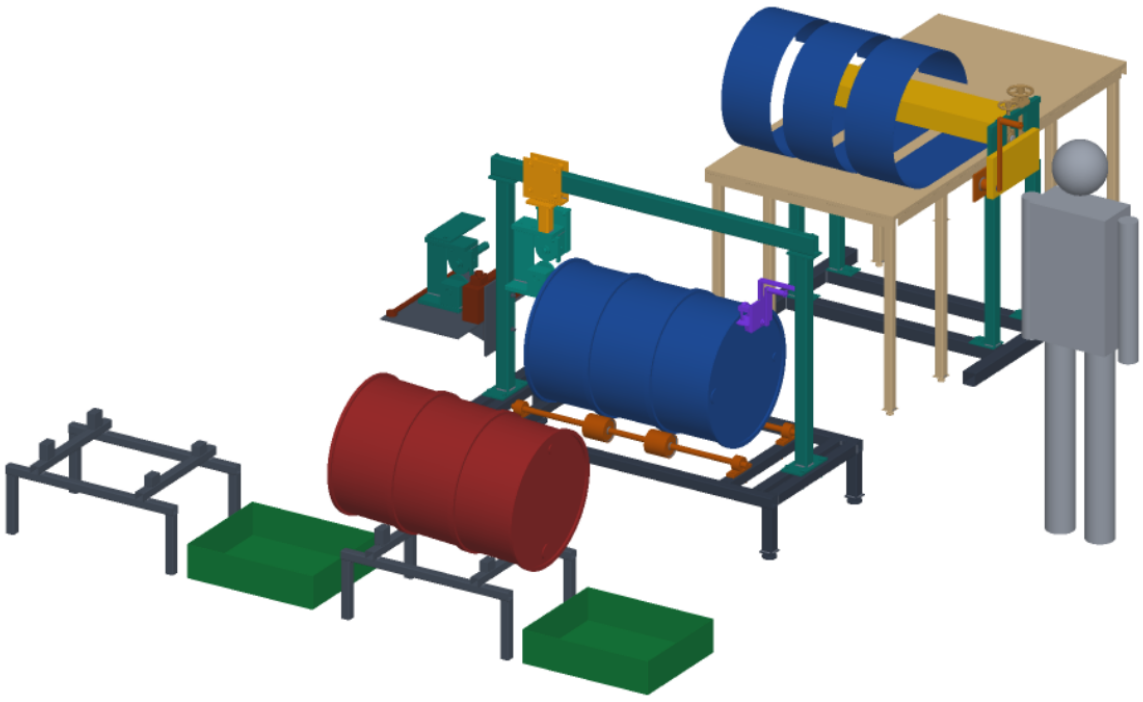
DrumPanel design precis

Turns empty oil drums into flat sheet by purging and cold-opening them instead of burning them.

How it works

DrumPanel is a hand-powered bench set in three stations that takes an empty 208 L oil drum to flat steel without fire. Two people run it; nothing needs electricity.

1. **Accept or reject.** Only a drum whose label shows it held a non-volatile oil is taken in. Anything else goes back.
2. **Drain and purge** (two drain and purge stands, so one drum soaks while the next is filled). The drum lies on a low stand that slopes 2.9 degrees toward its bungs, over a tray. Residue drains out of the bottom bung; the drum is then filled with water and a little detergent through the top bung, left 10 minutes, and drained. The water settles in a spare drum and is reused.
3. **Vapour check.** A pumped gas detector samples through both bungs. The drum passes only below 5 % of the lower explosive limit; otherwise it is purged again.
4. **Heads off** (cutting cradle). The drum rolls onto four rollers that carry it on its rolling hoops. A hand-cranked end cutter clamps over the chime: a knurled wheel drives it round while a hardened wheel cuts the head out just inside the chime. A torque arm against a rail post stops it turning with the drum. Both heads come out as flat discs about 552 mm across.
5. **Notch and slit.** A lever notching punch, slid into the open drum end, punches a notch through each chime and a slot through each hoop on the slit line, beside the drum's own welded seam, with up to about 18 swings of its ratchet lever for each. The drum brake is screwed on. A hand-cranked rotary shear head, hanging from a trolley on an overhead rail, is wound along the top of the drum: its upper disc is driven through a small chain drive whose crank turns high, clear of the drum, its lower disc runs inside, and its thin web runs in the cut behind them, so the drum is slit from end to end.
6. **Ring cuts.** The head is swung 90 degrees on its pivot and a second head, the ring head, is lowered through the slit and bolted to it 233 mm along the drum, its upper disc coupled to the first. With the brake off, the crank turns the drum on its rollers past both pairs of discs three times, making two cuts each time: 30 mm in from each end to take off the chime rings, and each side of each hoop to take out the hoop strips. What is left is three curled panels.
7. **Roll flat** (slip roll stand). The three panels are fed side by side, paint side up, through a hand-cranked slip roll: a pinch pair geared together drives them, and a bending roll behind bends them the other way just enough that they spring back flat. The bending roll is set on fine-pitch screws with dials and lock nuts: a trial pass on the first drum of each batch finds the dial reading; on every later drum the first panel is checked with a straightedge, the dial trimmed, and a panel out of tolerance passed again. The last 90 mm of each panel end is set flat by hand on a steel plate.
8. **Deburr.** Every cut edge is filed or scraped before the panels are stacked.



Seen from the front right and above, 24 deg elevation. Front: two drain and purge stands; middle: cutting cradle; back: slip roll. Grey figure: 1.75 m person for scale

Figure 1. The three stations: drain and purge stand (front), cutting cradle (middle), slip roll stand (back).

Components

Table 1. Components, in build order. Numbers match `bom/bom.csv`.

BOM	Component	Role
1	Drain and purge stands (2)	Welded frames with hardwood chocks that each hold a drum, full of water if need be, bung end low over a tray; two, so that one drum purges while another soaks
2	Drain trays (2)	Catch residue and purge water
3	Purge kit	Hose, bung adaptors, non-sparking bung wrench, settling drum fittings, detergent
4	Vapour check kit	Pumped LEL gas detector with its alarm at 5 % LEL, and bump-test gas
5	Cradle frame with levelling feet	Low welded frame; the feet also set the drum height under the rail
6	Cradle rollers, shafts and pillow blocks	Four rollers under the rolling hoops let the drum turn
7	Drum brake	Screw and pad that hold the drum still for the slit
8	Rail posts and rail beam	Portal over the cradle that carries the shear head
9	Trolley, swivel and drop bar	Carries the head along the rail; the pivot sets slit or ring position
10	Rotary shear head	Two hardened discs, driven upper and idle lower, in a frame whose web runs in the cut; a 2.33 to 1 chain drive in a closed case with a 155 mm crank
11	End cutter	Cuts each head out just inside the chime
12	Roll stand base	Wide base that keeps the slip roll from tipping
13	Side frames	Plates on legs that carry the three rolls
14	Rolls	Lower and upper pinch rolls and the bending roll, 60 mm
15	Bushes, slide blocks and screws	Bronze bushes; blocks and screws that set the pinch and the bend; dials and lock nuts on the fine-pitch bending screws
16	Pinch gears	Keep the pinch rolls turning together
17	Crank and chain drive	300 mm crank and a 3:1 chain to the lower roll
18	Roll guards	Nip guards with 8 mm slots, lid, chain and gear guards
19	In-feed and out-feed tables	Support the 1.8 m panels at the sheet line
20	Finishing tools and bench plate	Plate and mallet for the panel ends; deburring tools
21	Lever notching punch	C-frame punch with two stations that notches the chimes and slots the hoops on the slit line
22	Procedure sheets	Pictogram sheets for acceptance, purge, check, cutting, rolling and finishing
23	Fasteners, paint and welding consumables	Allowance
24	Personal protective equipment	Cut-resistant gloves, glasses, ear defenders, aprons

BOM	Component	Role
25	Ring head with spacer and coupling	A second shear head bolted to the first for two ring cuts at a time
26	Tool shelf	Shelf on the left rail post for the notching punch and the ring head

Key design choices

Choices 1 to 8 were decided on 2026-10-03 under Amish's pre-approval (DMP-DDR-001 and DMP-DDR-002); choices 9 and 10 are Amish's requirement decisions of 2026-10-03 (DMP-DDR-003). All are in the register DMP-DEC-001.

1. **Water purge, not air or steam.** Water displaces vapour and washes residue with no power or fuel; the water is reused.
2. **A bought pumped LEL detector, pass below 5 % LEL.** Half the usual 10 % hot-work limit, because a cold cut still has friction and a failed check costs only a second purge.
3. **Accept only drums labelled for non-volatile oils.** The purge is sized for oils, not fuels or solvents.
4. **Three panels, hoops cut out.** Straightening a rolling hoop would stretch its crown about 8 %; cutting the hoop strips out leaves three flat panels about 233 x 1,800 mm instead of one buckled sheet.
5. **One shear head for the slit, two for the ring cuts.** It swivels 90 degrees on its pivot; the slit is made with the drum still, the ring cuts with the drum turning and the ring head bolted on.
6. **Slip roll, not a three-roll pyramid.** A pyramid of three plain rolls cannot drive the panels by friction alone; the geared pinch pair grips them with a set force and the bending roll does the bending.
7. **Trial-pass setting.** The setting that leaves a panel flat changes with each batch of drum steel, and even from drum to drum, so the first panel of each batch is used to find it and the first panel of every drum is checked with a straightedge and the dial trimmed (STEM fix F1).
8. **Shared bench set.** One set serves a cooperative or a cluster of workshops.
9. **Throughput (R7).** A lever notching punch replaces the hacksaw notches and a ring head makes two ring cuts a pass, so two people turn about 2.20 drums an hour, against Amish's restated target of 2.5.
10. **Flatness (R5).** Fine-pitch bending screws with dials (0.025 mm a division) and lock nuts set the bending roll well inside the 0.29 mm allowance and hold it.

First-order numbers

Table 2. Key figures from DMP-CAL-001. Assumptions: drum steel yield 200 to 300 MPa, body 0.8 to 1.2 mm, heads 1.0 to 1.5 mm, shear strength 0.8 of a 360 MPa tensile strength.

Quantity	Value	Source
Output per drum	Three panels 233, 234 and 233 x 1,800 mm; two head discs 552 mm	[A1 to A3]
Flat area per drum	1.26 m² of panels and 0.48 m² of heads	[A4, A5]
Drum full of water	About 240 kg, filled and drained in place	[B2]
Purge	Fill 11 min at 20 L/min; drain 2.3 min; about 1.4 L left below the bung	[B3 to B5]
End cutter	Crank force 12 to 27 N; about 2 min a head with setting up	[C10 to C15, H7]
Shear head	Separating force up to 1,429 N; crank force 16 to 68 N for the slit, 32 to 136 N for two ring cuts	[D8 to D15r]
Notching punch	24.3 kN for a chime notch; lever force up to 131 N	[K1, K3]
Slip roll	Bending roll about 13.4 mm above the lower roll for 250 MPa, 1.0 mm steel; 9.3 to 19.3 mm over the range; set to 0.29 mm, 11.8 dial divisions	[E10, E11, E13, E14d]
Slip roll crank	20 N at 300 mm through a 3:1 chain; feed 1.9 m/min	[E23, E24]
Throughput	About 2.20 drums an hour for two people, with two drain stands (estimate; R7 asks 2.5)	[H5, H13]
Masses	Cradle with its tools about 196 kg, slip roll stand about 191 kg, each drain stand about 15 kg	[I1 to I3]
Cost	Value-engineering target: USD 3,000. Estimated cost of the constructable design: USD 3,566 (USD 566 over the target)	[J1 to J3]

Patent design-arounds

From the preliminary patent, trademark and prior-art screen (not legal advice):

- Stay manual; powered drum cutter-flatteners already exist. Every DrumPanel station is hand-cranked.
- Publish the purge and vapour check procedure openly with the design. It is in this precis, the build plan and the procedure sheets.
- Build on expired prior art such as the US3161952A drum-head cutter. The end cutter follows its principle: a wheel cutting the head just inside the chime.

Shared blocks

- Hand crank and chain drive: the slip roll's 300 mm crank with a #40 chain at 3:1 is drawn so it can later be swapped for the proposed hand capstan common block (SaltDrag, SiltHaul) if that block is adopted (DMP-DDR-001, D10).
- CalRig proof-load: used at TRL 4 for the drain stand, frame and post checks; nothing in the TRL 3 design depends on it.

Safety

Safety: An empty oil drum can still hold flammable vapour. Never cut, grind, weld or heat a drum until the purge is done and the vapour check passes below 5 % LEL; repeat the check if a drum stands overnight.

Safety: Do not process drums that held fuel, solvent, chemicals or unknown contents, or that have no readable label.

Safety: Cut steel edges are sharp. Wear cut-resistant gloves and eye protection for every task that handles cut steel, and deburr every edge before panels are stacked or carried.

Safety: The shear head, ring head, end cutter and slip roll have in-running nips, and the slip roll and the shear head have chains. Keep every guard fitted (a tool is needed to remove them), keep one person on a crank and nobody else's hands near the work, and never reach past a guard.

Safety: A drum full of water weighs about 240 kg. Fill and drain it on the stand; never lift or tip a full drum. Lift anything over 25 kg with two people.

Safety: Purge water carries oil. Settle it, skim the oil into a closed container for a recycler, and never pour it on the ground or into a drain.

This design is published as an open engineering reference. It is not certified equipment.

Next steps

TRL 3 is complete on paper (DMP-CAL-001, DMP-BLD-001). The recommended next step, for Amish to authorise when the portfolio phase allows it, is TRL 4: approach the first co-design candidate, build the bench set to DMP-BLD-001, and run the first checks in its section 5, starting with the vapour check procedure and the slip roll trial pass (R2, R5).